



POLITECNICO
MILANO 1863 | DIPARTIMENTO
DI MECCANICA

Lab- Human-In-The-Loop Dynamic Driving Simulator

Mehmet Atceter
11012941

Matteo Cigada
10767277

Lorenzo Crosta
10794858

Federico D'Agostini
10788635

Michele Vignati,
Federico Maria Ballo

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2. Comparison with offline simulation
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Description of the structure of the presentation

- Aim of the lab
 - Analyze the effect of a real drive inside a vehicle simulator (human-in-the-loop)
 - Different drivers produce different results
 - A human driver produces different results than the off-line simulation
 - Design of special maneuver and scenario to highlight than instance
 - Learn how to use the specific software to design and perform the simulation

Intro

00

00: Intro: definition of the maneuver

Different lane changes are tested, for different speed and emergency levels.



- Definition of the maneuver: lane change has been selected
- Different difficulty and speed for the same maneuver
 - **Slow emergency lane change**
 - Obstacle: fixed object (construction works)
 - Speed: slow (50km/h)
 - Difficulty: blind obstacle
 - Aimed to be performed at car limit
 - **Standard lane change**
 - Obstacle: vehicle moving in the same direction
 - Speed: fast (90km/h)
 - Difficulty: no difficulty, standard overtake
 - Aimed to be performed in the safest way possible
 - **Very fast lane change**
 - Obstacle: vehicle coming towards the driver
 - Speed: medium (60km/h)
 - Difficulty: driver must follow a truck, therefore the incoming vehicle is completely blind until the last moment
 - Aimed to be performed at car limit (each driver reflexes will be influential)

Real time simulation

01

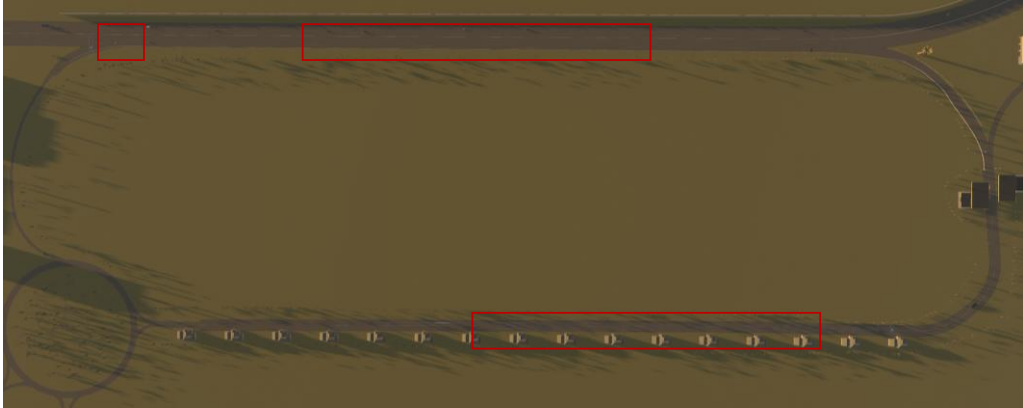


Real time simulation: Scenario

01.1

01.1: Real time simulation: Scenario

Environment



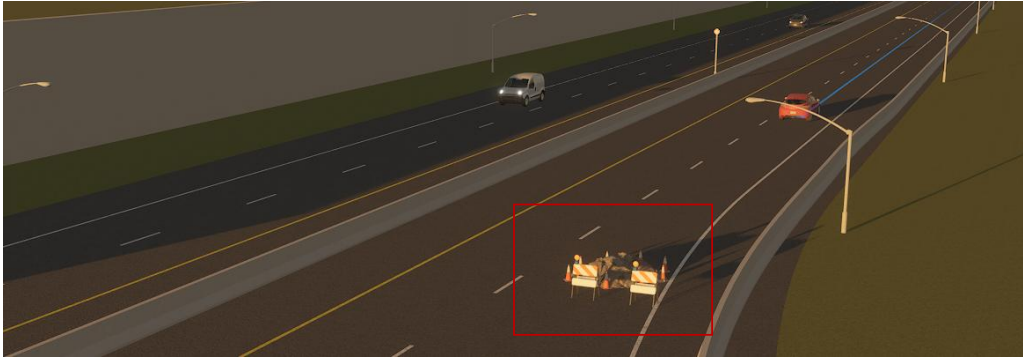
Description of the choice

1. Fixed blind obstacle
 1. In the entry lane of the highway
2. Long highway straight
 1. The fast overtake is performed there
3. Long extra urban road
 1. The street where the final obstacle is placed
 2. There is enough space to turn right or left once the obstacle is visible
 3. Repetitive sequence of houses and trees makes it hard for the driver to have a reference

Other objects have been placed to increase the realism of the simulation.

01.1: Real time simulation: Scenario

First obstacle: Fixed obstacle



- The object is (partially) hidden by the jersey on the right and placed at the entrance of the highway. The driver is focused on trying to enter in the highway, at the same time it sees the obstacle and tries to avoid it.

01.1: Real time simulation: Scenario

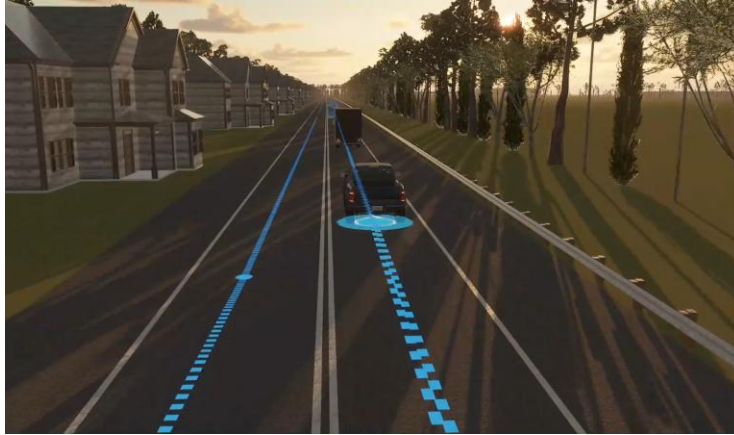
Second obstacle: high speed overtake



The car is going at fixed speed at 80km/h, while the driver is going at 90km/h. The objective is to perform a standard overtake at high speed in the safest way possible.

01.1: Real time simulation: Scenario

Third obstacle: Incoming vehicle



The truck act as a mask going at 60 km/h that hides the car coming in the opposite lane, it moves at the last moment to avoid the car. The driver sees the car in the opposite lane at the last second and needs to react fast to avoid the incoming danger.

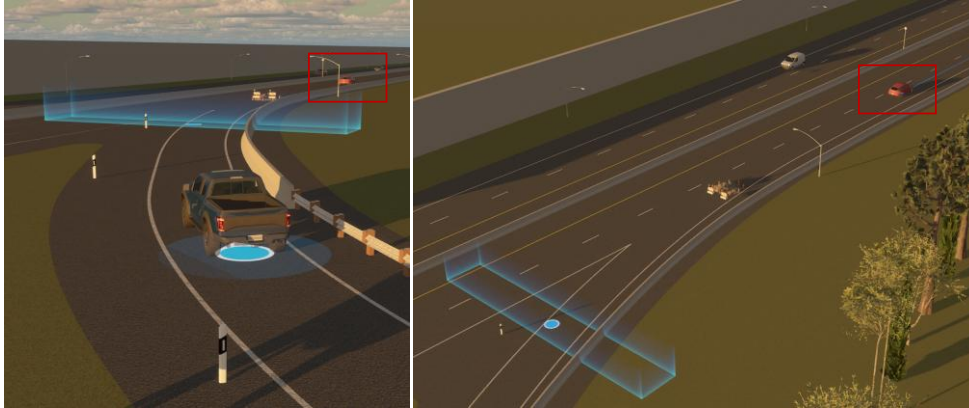


Real time simulation: Traffic implementation

01.2

01.2: Real time simulation: Traffic implementation

Second obstacle: moving vehicle to be overtaken



- The ego vehicle activates a trigger that causes the orange vehicle to start moving and accelerate up to 80 km/h

01.2: Real time simulation: Traffic implementation

Third obstacle: Truck and incoming vehicle



Sequence of 3 events

1. The ego vehicle activates **first trigger** that causes the truck (yellow rectangle) to start moving and accelerate up to 60 km/h
2. The truck activates the **second trigger** that causes the pink vehicle (red square) to start moving and accelerate up to 60 km/h
3. When the relative distance between the truck and the pink vehicle gets around 50m, the **third trigger** activates and causes the truck to change lane, making the moving obstacle visible to the driver of the ego vehicle. The truck has a double function: mask the obstacle coming towards the vehicle and force the driver to keep the distance as constant as possible.

01.2: Real time simulation: Traffic implementation

Third obstacle: Truck and incoming vehicle



Sequence of 3 events

1. The ego vehicle activates **first trigger** that causes the truck (yellow rectangle) to start moving and accelerate up to 60 km/h
2. The truck activates the **second trigger** that causes the pink vehicle (red square) to start moving and accelerate up to 60 km/h
3. When the relative distance between the truck and the pink vehicle gets around 50m, the **third trigger** activates and causes the truck to change lane, making the moving obstacle visible to the driver of the ego vehicle. The truck has a double function: mask the obstacle coming towards the vehicle and force the driver to keep the distance as constant as possible.

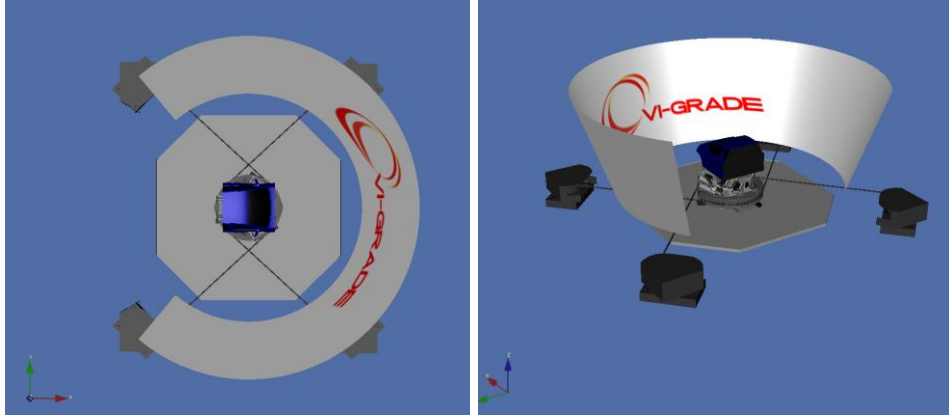


Real time simulation: Cueing

01.3

01.3: Real time simulation: Cueing

The issue



- Why cueing:
 - real accelerations/velocities not directly reproducible
 - limited simulator displacement
 - need filtering/scaling of motion signals

01.3: Real time simulation: Cueing

First iteration tested in DriveSim using offline simulations, final iteration tuned during the online calibration runs

First Iteration		
Group	Type	Value
Axis of Rotation (AoR)	-	Automatic AoR
	-	Yaw
	-	Pitch
	-	Roll
	-	Tilt Rotation Center
Longitudinal	Filter	Acceleration HP / LP / Tilt LP
	Gain	Acceleration Gain / Non Linear / Tilt Gain
	Feature	Balance / Inversion Avoidance
	Feature	Offset / Hexapod Supply
Lateral	Filter	Acceleration HP / LP / Tilt LP
	Gain	Acceleration Gain / Non Linear / Tilt Gain
	Feature	Inversion Avoidance / Offset / Hexapod Supply
	Feature	Velocity HP / Velocity LP
Yaw	Gain	Velocity Gain / Non Linear Gain
	Feature	Inversion Avoidance / Offset / Hexapod Supply
Yaw Dynamics	Filter	vy / HP / P / HP
	Gain	vy / P / B
Pitch	Filter	Velocity HP / Velocity LP
	Gain	Velocity Gain
Roll	Filter	Velocity HP / Velocity LP
	Gain	Velocity Gain
Vertical	Feature	Axis Inclination Angle
	Filter	Acceleration HP / HP HF / LP
	Gain	Acceleration Gain / Acceleration HF
	Feature	Inversion Avoidance / Offset
	Feature	Unchecked (0) / 0 m



Final Iteration		
Group	Type	Value
Axis of Rotation (AoR)	-	Automatic AoR
	-	Yaw
	-	Pitch
	-	Roll
	-	Tilt Rotation Center
Longitudinal	Filter	Acceleration HP / LP / Tilt LP
	Gain	Acceleration Gain / Non Linear / Tilt Gain
	Feature	Balance / Inversion Avoidance
	Feature	Offset / Hexapod Supply
Lateral	Filter	Acceleration HP / LP / Tilt LP
	Gain	Acceleration Gain / Non Linear / Tilt Gain
	Feature	Inversion Avoidance / Offset / Hexapod Supply
	Feature	Velocity HP / Velocity LP
Yaw	Gain	Velocity Gain / Non Linear Gain
	Feature	Inversion Avoidance / Offset / Hexapod Supply
Yaw Dynamics	Filter	vy / HP / P / HP
	Gain	vy / P / B
Pitch	Filter	Velocity HP / Velocity LP
	Gain	Velocity Gain
Roll	Filter	Velocity HP / Velocity LP
	Gain	Velocity Gain
Vertical	Feature	Axis Inclination Angle
	Filter	Acceleration HP / HP HF / LP
	Gain	Acceleration Gain / Acceleration HF
	Feature	Inversion Avoidance / Offset
	Feature	Unchecked (0) / 0 m

- Workflow:
 - first iteration from offline simulations
 - final iteration from online calibration runs
 - corrections suggested by Paolo
 - tuning based on real-time simulator behaviour
- Main objective:
 - not simply stronger/weaker cueing
 - rebalance longitudinal/lateral/rotational channels
- Final target:
 - stronger longitudinal perception
 - softer lateral response in lane changes
 - smoother yaw and pitch cues

01.3: Real time simulation: Cueing

Final iteration tuned during the online calibration runs

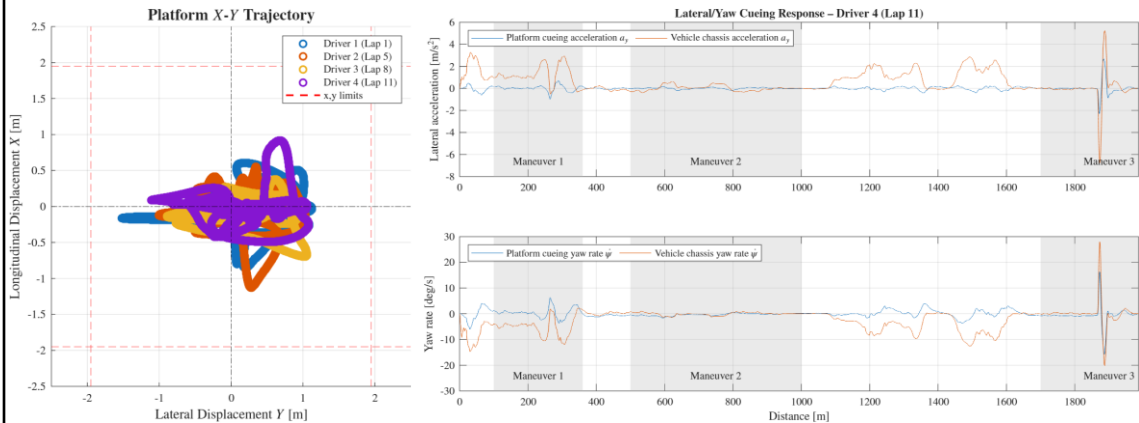
Comparison				
Group	Type	Parameter	First Iteration	Final Iteration
Axis of Rotation (AoR)	-	Automatic AoR	Checked	Unchecked
Longitudinal	Filter	Acceleration HP / LP / Tilt LP	0.55 Hz / 95 Hz / 0.25 Hz	0.25 Hz / 95 Hz / 0.25 Hz
Longitudinal	Gain	Acceleration Gain / Non Linear / Tilt Gain	3.5 / 0.1 / 0.2	3.5 / 0.1 / 0.4
Lateral	Gain	Acceleration Gain / Non Linear / Tilt Gain	5.5 / 0.1 / 0.1	4.5 / 0.1 / 0.1
Lateral	Feature	Inversion Avoidance / Offset / Hexapod Supply	Checked (0) / 0.2 m / 0	Unchecked (0) / 0.2 m / 0
Yaw	Filter	Velocity HP / Velocity LP	0.35 Hz / 95 Hz	0.1 Hz / 95 Hz
Yaw	Gain	Velocity Gain / Non Linear Gain	1 / 0	0.7 / 0
Pitch	Gain	Velocity Gain	1	0.5

- Longitudinal acceleration HP frequency ↓
 - more medium-low frequency content
 - more continuous longitudinal cue
 - better speed/acceleration perception
- Longitudinal tilt gain ↑
 - sustained longitudinal acceleration via pitch tilt
 - gravity component perceived as acceleration/braking
 - stronger acceleration/deceleration cue
- Pitch velocity gain ↓
 - avoid excessive pitch motion
 - compensate increased longitudinal tilt
 - better pitch/longitudinal balance
- Lateral acceleration gain ↓
 - softer lateral cue
 - avoid over-cueing in sharp lane changes
 - better balance with yaw and visual feedback
- Yaw velocity HP frequency ↓
 - slower yaw components retained
 - less impulsive yaw cue

- more continuous vehicle rotation perception
- Yaw velocity gain ↓
 - lower yaw peaks
 - smoother yaw response
 - less aggressive but more persistent cue

01.3: Real time simulation: Cueing

Parameters tuned during the calibration runs



- X-Y platform displacement:
 - limits read from .vicue
 - used as workspace boundaries
 - no evident horizontal saturation
- Lateral acceleration cueing:
 - Maneuver 1: main trend reproduced
 - Maneuver 2: weaker match, less critical
 - Maneuver 3: strongest transient reproduced
 - captures main events, not perfect shape matching
- Yaw-rate cueing:
 - better than lateral acceleration
 - clearer timing/shape agreement
 - Maneuver 1 and 3: main rotational events captured
 - Maneuver 2: lower yaw activity, limited response
 - yaw channel preserves rotational content more effectively



Real time simulation: Driving the DriSMI

01.4

01.4: Real time simulation: Driving the DriSMI

Runsheets

run number #	driver	maneuver 1 (fixed obstacle)	maneuver 2 (simple overtake)	maneuver 3 (moving obstacle avoidance)	additional notes	vehicle used	cueing	sim type	csv results	preprocessed filename
1	MA	ok	overtook too early	overtook the truck	invalid results	pickup	DriSMI cueing	static	NO ?	NOT PREPROCESSED
2	MA	ok	overtook too early	collision	stopped early	pickup	DriSMI cueing	static	YES	
3	MA	ok	ok, speed not constant	collision		pickup	DriSMI cueing	static	YES	
4	MA	ok	ok, speed not constant	ok		pickup	DriSMI cueing	static	YES	
5	MA				gear changes test	pickup	DriSMI cueing	static	YES	
6	MA				gear changes test	pickup	DriSMI cueing	static	YES	
7	MA	ok	ok, speed not constant	ok	changes in cueing	sedan	custom cueing: hint: in the next run decrease roll gain to 0.5 to make it closer to lateral one; remove the flag in lateral; match gain on yaw closer to lateral acceleration one, static	static	YES	NOT PREPROCESSED
8	MA	ok	ok, speed not constant	ok	changes in cueing	sedan	custom cueing: hint: in the next run decrease lateral gain to 5 and increase longitudinal one to 0,4	static	YES	Driver1_Debug1
9	MA	ok	ok, speed not constant	ok, followed truck too early		sedan	custom cueing	dynamic	YES	Driver1_Lap1
10	MA	ok	ok, speed not constant	ok, followed truck too early	changes in cueing	sedan	custom cueing: hint: in the next run reduce lateral gain to 4.5 and set initial speed to 10 km/h	dynamic	YES	Driver1_Lap2
11	FD	ok	ok, speed constant	ok, followed truck too early + too far from the truck		sedan	custom cueing	dynamic	YES	Driver2_Lap3
12	FD	ok, speed too low	ok, speed constant	ok		sedan	custom cueing	dynamic	YES	Driver2_Lap4
13	FD	ok	ok, speed constant	collision, LC indicated to go right		sedan	custom cueing	dynamic	YES	Driver2_Lap5
14	LC	ok	ok, speed constant	ok, too far from the truck		sedan	custom cueing	dynamic	YES	Driver3_Lap6
15	LC	ok	ok, speed not constant	collision, MC indicated to go left		sedan	custom cueing	dynamic	YES	Driver3_Lap7
16	LC	ok	ok, speed not constant	ok, MC indicated to go left		sedan	custom cueing	dynamic	YES	Driver3_Lap8
17	MC	ok	ok, speed constant	ok followed the truck too early + a little bit too early		sedan	custom cueing	dynamic	YES	Driver4_Lap9
18	MC	ok	ok, speed almost constant	ok FD indicated to the right AND HE SUCCEEDED		sedan	custom cueing	dynamic	YES	Driver4_Lap10
19	MC	ok	ok, speed constant	ok FD indicated to the left, he found it easier than before		sedan	custom cueing	dynamic	YES	Driver4_Lap11

- This runsheet summarizes the real-time simulation sessions performed in the DriSMI. The pickup vehicle was used at first, but it was later replaced by the sedan because gear-shift issues were observed during the simulation. For each run, we tracked the driver, the maneuver outcomes, the cueing setup, and the output data. In the first run, drivers were free to choose the path, while in the following runs the path was prescribed.
- First run each driver was free to choose where to go, in the other two he was told where to go
- This runsheet summarizes all the real-time simulation sessions performed in the DriSMI.
- For each run, we recorded:
 - the driver
 - the outcome of the three maneuvers
 - additional remarks

- the vehicle configuration
 - the cueing setup
 - the simulation type
 - the corresponding output files.
- In the first run, each driver was free to choose the path to follow, while in the following two runs the path was prescribed.
 - This table was used to keep track of valid runs, identify issues such as collisions or early maneuvers, and select the data for the subsequent analysis.

Offline simulation

02



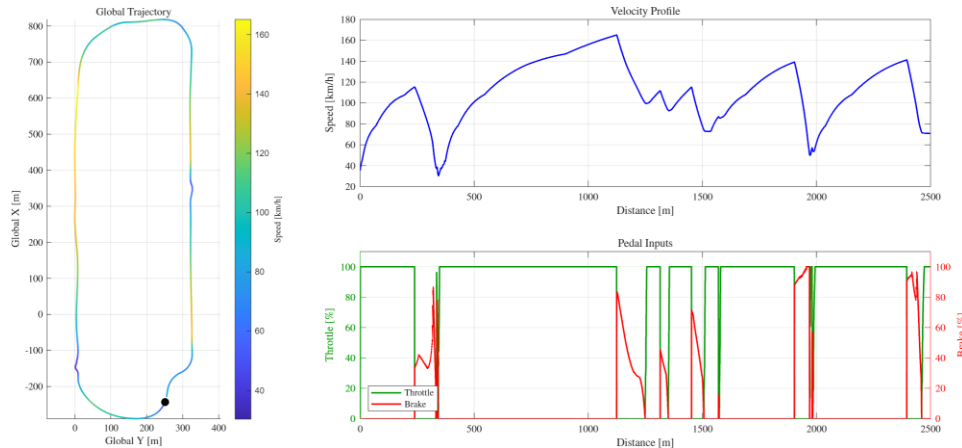
Offline simulation: Custom trajectory – max performance

02.1

- First step: draw «by points» a custom trajectory coherent with the choice of the 3 maneuvers (first obstacle represented by a sharp trajectory change, second obstacle represented by a smoother and wider one)

02.1: Offline simulation: Custom trajectory – max performance

Max performance simulation along a custom trajectory



- Max performance event performed on the custom trajectory to have an idea of the maximum speed at which the three maneuvers can be performed
- This result refers to a maximum-performance simulation carried out along a custom trajectory defined offline. Since the path is predefined, the vehicle control is optimized only to achieve the highest possible performance along that route while respecting the physical limits of the vehicle. This case is therefore used as an upper-bound reference for the maneuver.

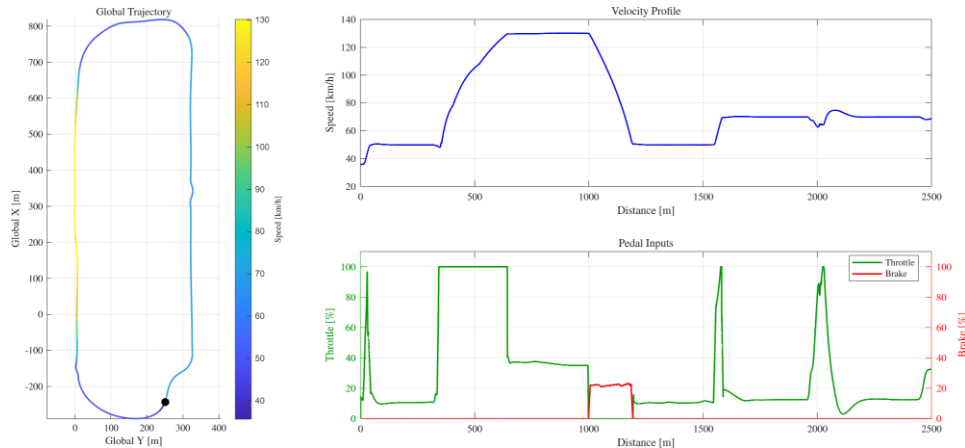


Offline simulation: Custom trajectory – custom speed profile

02.2

02.2: Offline simulation: Custom trajectory – custom speed profile

Custom speed profile simulation along a custom trajectory



- Custom speed profile is imposed based on the information provided by the max performance event
- Low speed in the connection between Maneuver 2 and 3 not to have issue with the cueing
- This simulation was used to define the first iteration of the cueing
- On the left, we can see the custom trajectory defined offline, while on the right we have the corresponding velocity profile and pedal inputs obtained with the imposed custom speed profile. In this case, the custom speed profile is defined based on the information obtained from the max-performance event. In particular, a lower speed is imposed in the transition between Maneuver 2 and Maneuver 3 in order to avoid issues with the cueing system. This simulation was then used to define the first iteration of the cueing strategy.



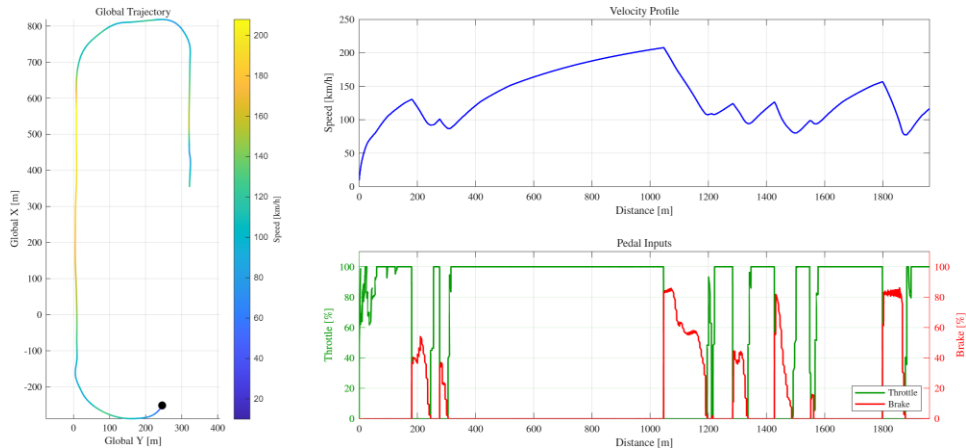
Offline simulation: Online trajectory: max performance

02.3

- Second step : take the trajectory from the online simulations and repeat the two steps

02.3: Offline simulation: Online trajectory – max performance

Max performance simulation along the online simulation trajectory



- Max performance event performed on the actual trajectory to have an idea of the maximum speed at which the trajectory covered by the drivers can be performed. This will be used to make a comparison between online and offline simulations

On the left, we can see the trajectory recorded from one of the drivers during the real-time simulation, while on the right we have the corresponding velocity profile and pedal inputs obtained in the offline max-performance run. In this case, the vehicle is driven at the maximum achievable performance along the actual path followed by the driver, while still respecting the physical limits of the vehicle. This result is useful to estimate the highest speed theoretically achievable on a trajectory generated during the online simulation, and it provides a reference for comparison with the driver's real performance.



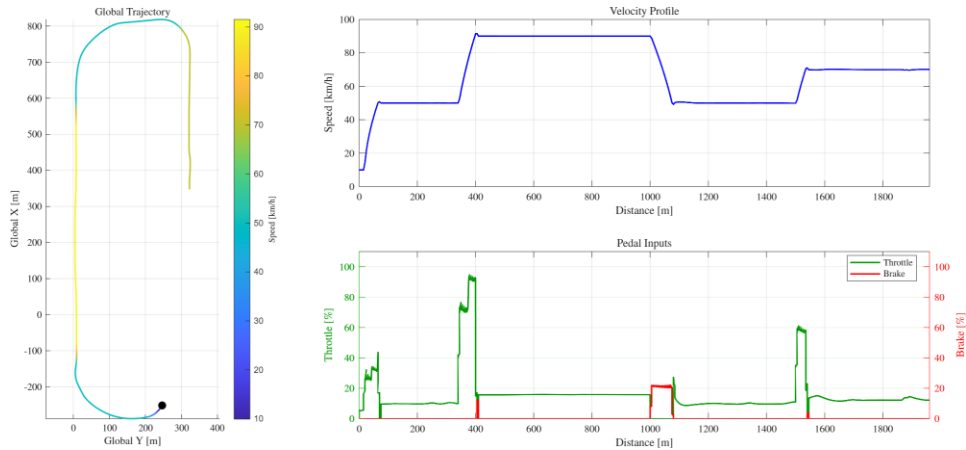
Offline simulation: Online trajectory: custom speed profile

02.4

- Second step : take the trajectory from the online simulations and repeat the two steps

02.4: Offline simulation: Online trajectory – custom speed profile

Custom speed profile simulation along the online simulation trajectory



In this case, the same trajectory recorded from one of the drivers during the real-time simulation is used again, but instead of applying a max-performance strategy, a custom speed profile is imposed along the path. This leads to a smoother and more controlled maneuver, with less aggressive speed variations and pedal inputs. The result is useful to compare a more realistic driving strategy with both the driver's actual behavior and the max-performance run on the same online trajectory.

Results

03



Results: Comparison between drivers

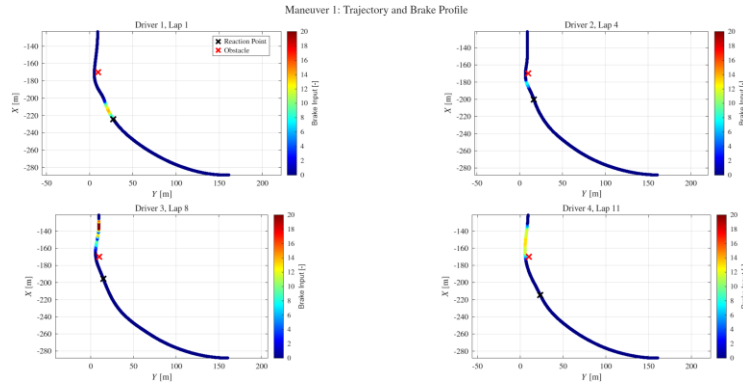
03.1

03.1: Results: Comparison between drivers

Maneuver 1

Reaction point:

- $\delta_{SW} > 25^\circ$
or
- $BRK_{\%} > 5\%$

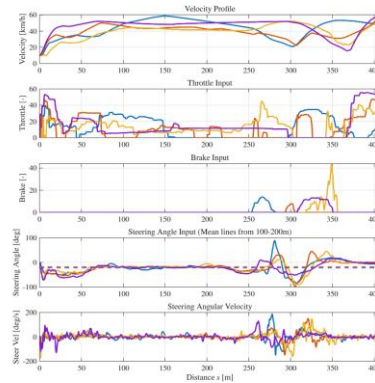
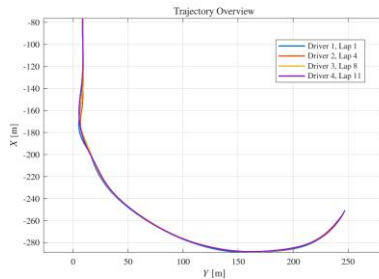


Driver, Lap	Reactivity & Distraction			
	Pre-Reaction Speed Std [km/h]	Pre-Reaction Speed MAE [km/h]	First Reaction Input	Reaction s [m]
Driver 1, Lap 1	4.39	4.11	Brake First	255.96
Driver 2, Lap 4	3.15	8.22	Steer First	278.64
Driver 3, Lap 8	2.12	7.34	Steer First	283.68
Driver 4, Lap 11	0.92	0.81	Steer First	262.97

- Drivers had 2 tasks
 - Avoid obstacle
 - Keep the speed constant → driver not exclusively focused on the obstacle
- First two columns: how good is the driver at keeping the speed constant
 - Driver 4 dedicating a lot of attention to the speed task
 - Drivers 2,3 more focused on the obstacle
- Description of the diagram
 - Brake input colorbar
 - Definition of the first input → Driver 1 only one applying brake as first input
 - Drivers 1,2 brake before obstacle

03.1: Results: Comparison between drivers

Maneuver 1

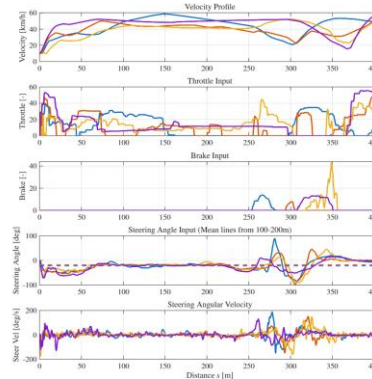
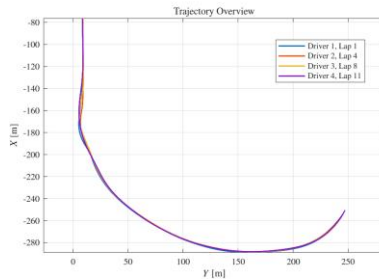


Driver, Lap	Peak Response KPI		
	Max Steering Rate [deg/s]	Max Lateral Acceleration [g]	Max Yaw Rate [deg/s]
Driver 1, Lap 1	187.67	0.24	14.97
Driver 2, Lap 4	147.92	0.18	14.27
Driver 3, Lap 8	161.17	0.53	21.28
Driver 4, Lap 11	130.30	0.30	11.95

- Absolute KPIs describing the severity of Maneuver 1
 - Max steering rate: driver input
 - Max lateral acceleration & Max Yaw rate: vehicle response
- Main interpretation:
 - strongest steering input does not automatically produce strongest vehicle response
 - Driver 1 has highest max steering speed but not strongest vehicle response
 - Driver 1, 2 max ay lower compared to Drivers 3, 4 → longitudinal dynamics, not purely lateral
 - response depends on speed, trajectory curvature, vehicle state and transient dynamics (remember drivers 1, 2 braked during the maneuver)

03.1: Results: Comparison between drivers

Maneuver 1



Driver, Lap	M1/M2 Amplification Factors		
	Max SteeringRate (M1/M2)	Max Lateral Acceleration (M1/M2)	Max Yaw Rate (M1/M2)
Driver 1, Lap 1	2.10	1.20	2.91
Driver 2, Lap 4	3.74	1.05	3.46
Driver 3, Lap 8	3.99	3.93	5.74
Driver 4, Lap 11	4.28	4.95	8.35

- Key idea:
 - amplification factors describe relative change from standard overtake to emergency lane change
 - not the same information as absolute peak values
- General trend:
 - all drivers increase steering rate and yaw rate
 - Maneuver 1 is clearly more abrupt and more rotational than Maneuver 2
- Important distinction:
 - peak KPIs → absolute severity in Maneuver 1
 - amplification factors → how much each driver changes behaviour with respect to Maneuver 2
- Critical note:
 - amplification ratios depend on the Maneuver 2 reference value
 - high ratio does not always mean highest absolute severity → for Driver 1,2 the amplification factor close to 1 simply means that maneuver 1 was performed as smooth, Driver 4 reaction more panicked
- Overall takeaway:
 - the most relevant result is not only braking vs no braking
 - the key point is the different conversion from steering input to vehicle

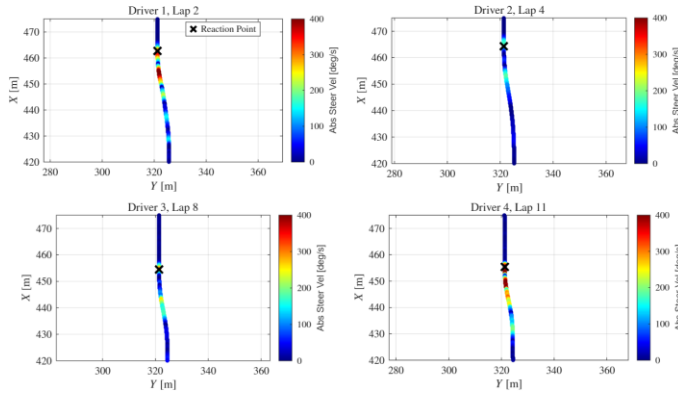
response

- Driver 1–2: more input/yaw amplification, limited lateral load increase
- Driver 3–4: stronger full dynamic amplification

03.1: Results: Comparison between drivers

Maneuver 3: correctly performed

Maneuver 3: Trajectory and Absolute Steering Velocity



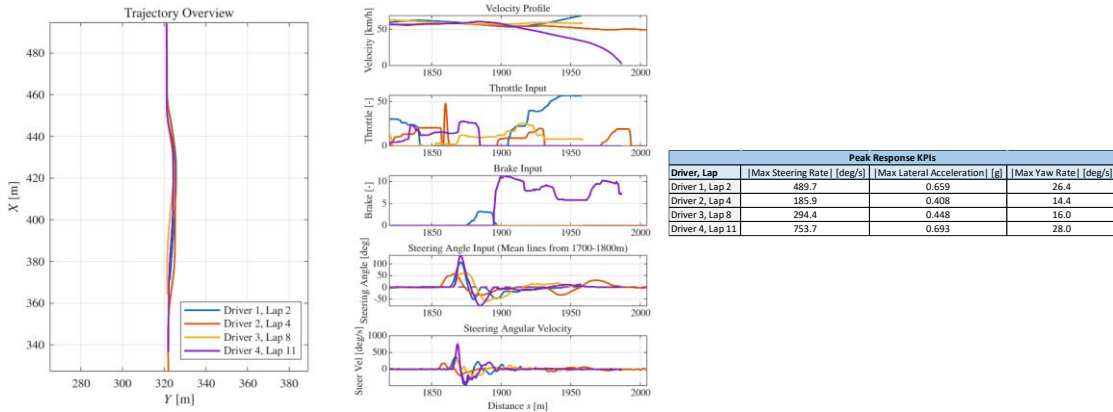
First Reaction		
	First Reaction Input	Reaction s [m]
Driver 1, Lap 2	Steer First	1866
Driver 2, Lap 4	Steer First	1859
Driver 3, Lap 8	Steer First	1868
Driver 4, Lap 11	Steer First	1867

First reaction input

- Everyone steer first
- The reaction distance, considering that the maneuver is performed between three moving vehicle, is similar
- Maybe also too far from the truck

03.1: Results: Comparison between drivers

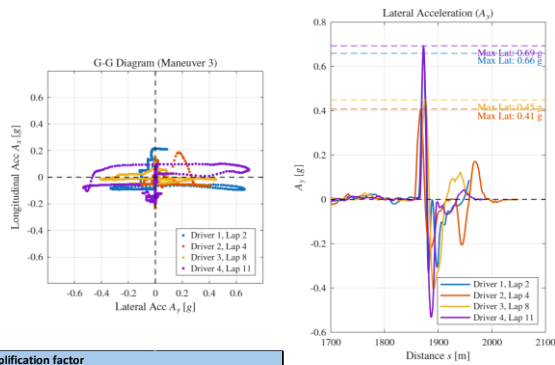
Maneuver 3 : correctly performed



- Throttle
 - Driver 3 and 4 perform the emergency steering maneuver while accelerating
- Brake
 - No driver use brake to avoid the obstacle
- Steering angle
 - Driver 1 and 2 use a considerably higher steering angle and a considerably higher steering velocity
 - Driver 3 starts steering considerably earlier
- Peak yaw rate
 - The yaw rate, for the car analyzed follows the peak steering angle trend

03.1: Results: Comparison between drivers

Maneuver 3: correctly performed



M3/M2 amplification factor			
Driver, Lap	Max SteeringRate (M3/M2)	Max Lateral Acceleration (M3/M2)	Max Yaw Rate (M3/M2)
Driver 1, Lap 2	7.76	5.53	7.86
Driver 2, Lap 4	4.70	2.38	3.49
Driver 3, Lap 8	7.29	3.31	4.31
Driver 4, Lap 11	24.74	11.52	19.54

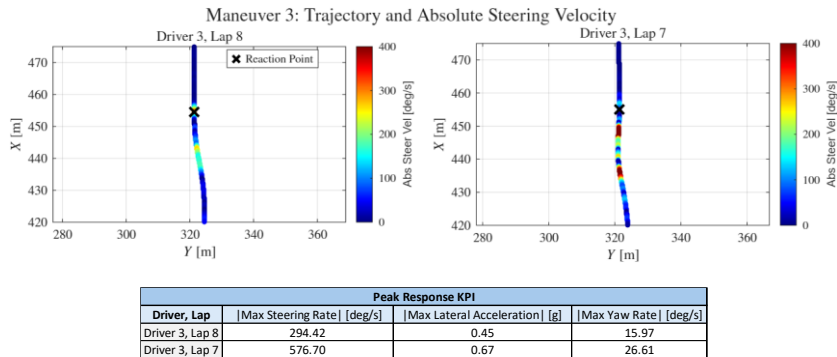
This maneuver is considerably different compared to our reference maneuver.

Weakness of the analysis:

- Different speed
- Different scenario
- Different trajectory

03.1: Results: Comparison between drivers

Maneuver 3: collision with the car



Why are these laps selected:

- Same driver
- Same direction
- Different results

What are the critical aspects of the analysis:

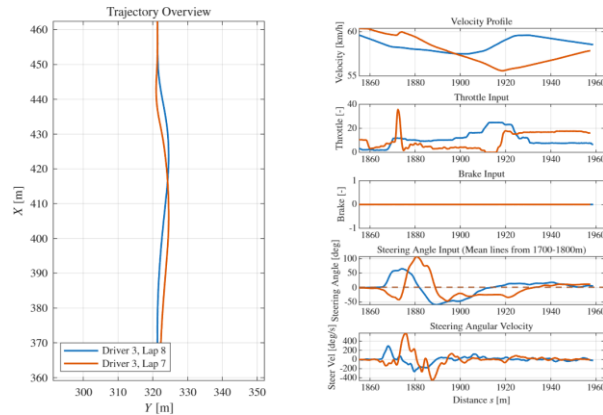
- Absolute position might not be the correct indicator, distance from the car in front and instant when the truck moves should be used as values.
- Also the timing for the direction indication is not well and precisely controlled.

Trajectory plot:

- The second trajectory, where the collision happens, is late compared to the first one
- In the second trajectory the driver slightly turn right before turning left, he probably assumed incorrectly that the indication would have been to go right.

03.1: Results: Comparison between drivers

Maneuver 3: collision with the car

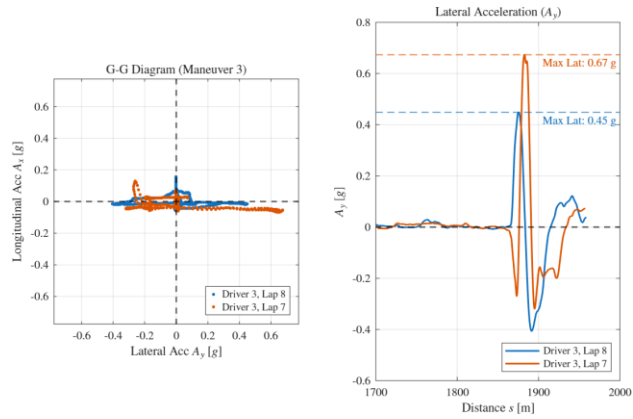


Differences:

- Throttle
 - Spike before the proper maneuver
- Brake input
 - No brake in both maneuver
- Steer angle
 - Collision
 - Turn right, then left after the call
 - Higher steering input
 - Higher steering speed

03.1: Results: Comparison between drivers

Maneuver 3: collision with the car



- Considerably higher lateral accelerations in the case where collision happens

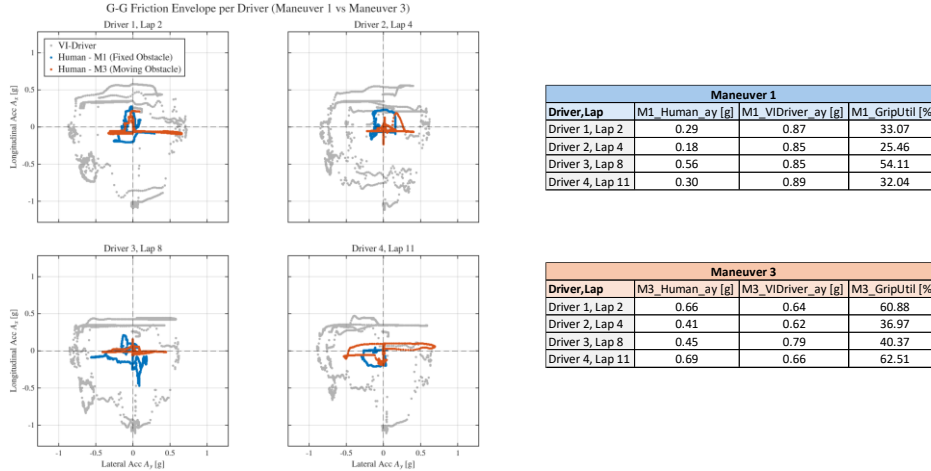


Results: Comparison with offline simulation

03.2

03.2: Results: Comparison with offline simulation

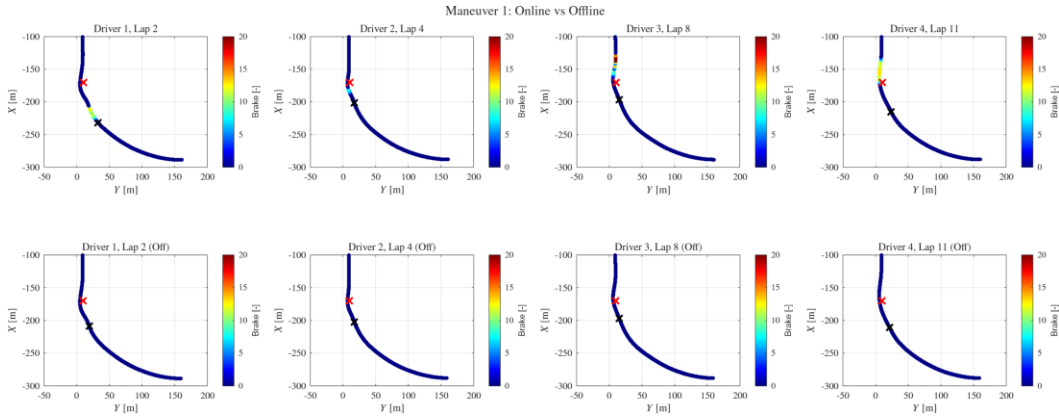
Comparison with the max performance simulation along the online simulation trajectory



- The G-G comparison shows that the human drivers remain well inside the VI-driver max-performance reference. Grip utilization is computed from the combined acceleration demand, $(\sqrt{a_x^2 + a_y^2})$, rather than from lateral acceleration alone.
- In Maneuver 1, grip utilization remains low for all drivers, ranging from about **25%** for Driver 2 to **54%** for Driver 3, confirming that the fixed-obstacle avoidance is far from the vehicle limit.
- Maneuver 3 is more demanding, especially for Driver 1 and Driver 4, which reach about **61–63%** grip utilization.
- However, even in the most critical cases, the drivers do not fully exploit the available adhesion margin.
- This suggests that the online maneuvers are mainly limited by driver perception, reaction strategy and simulator interaction rather than by the physical grip limit of the vehicle.

03.2: Results: Comparison with offline simulation

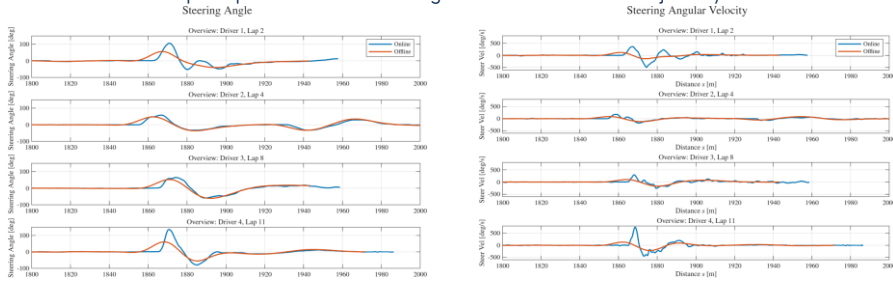
Comparison with the custom speed profile simulation along the online simulation trajectory



The offline simulations performed with our trajectories. During the first manoeuvre some of the driver tried to brake and then steer, because it may be the first reaction when there is an obstacle on the road and the car is moving at low speed and trying to enter in a highway.

03.2: Results: Comparison with offline simulation

Comparison with the custom speed profile simulation along the online simulation trajectory



Maneuver 3		
Driver	Steer angle on/Steer angle off	Steer vel on/Steer vel off
1	1,96	3,05
2	1,23	2,03
3	1,24	2,92
4	2,28	5,74

The offline simulations performed with our trajectories. During the third manoeuvre all the drivers tried to steer with a higher steering angular velocity with respect to the offline simulation. This behaviour is due to the unexpected danger that is coming towards the driver, so it tries to react fast to avoid the obstacle. Driver 2 and 3 had a behaviour like the one of the offline simulation, while Drivers 1 and 4 inserted a stronger steering input.



Results: Subjective analysis

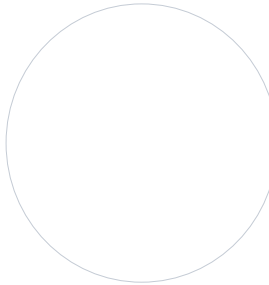
03.3

03.3: Results: Subjective analysis

Driver sensations and comments after the driving simulation

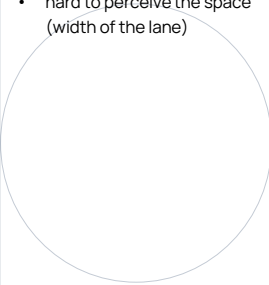
Driver 1:
Mehmet Atceter

- Longitudinal acceleration is not correctly perceived



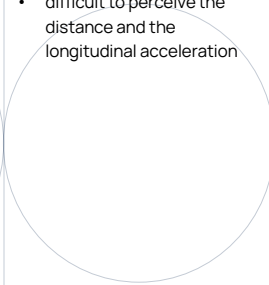
Driver 2:
Federico D'Agostini

- speed not correctly perceived
- hard to perceive the space (width of the lane)



Driver 3:
Lorenzo Francesco Crosta

- difficult to control the speed
- difficult to perceive the distance and the longitudinal acceleration



Driver 4:
Matteo Giuseppe Cigada

- longitudinal acceleration is hard to perceive
- first obstacle is not blind



The feedback collected after the driving simulation showed repeated difficulties in correctly feeling speed, longitudinal acceleration, and the available space during the maneuver. Even though the cueing strategy was adjusted to improve the longitudinal behaviour of the simulator, the final feedback still showed that longitudinal acceleration was not felt in a satisfactory way. This means that, despite the changes made in the cueing procedure, some limits were still present in how realistic the longitudinal sensation felt to the drivers.

03.4: Results: Possible improvements

Refinements and corrections in the approach that may be implemented

- Maneuver 1:
 - Fix the trigger for the first vehicle (it needs to start accelerating sooner / need to prevent the drivers from braking after Maneuver 1 if not needed by the maneuver itself)
- Maneuver 3:
 - Do not stop immediately after avoiding the collision with the incoming vehicle (need to isolate brake input related to emergency maneuver from the simulation ending)
- Whole test:
 - Modify longitudinal cueing settings

Based on both the simulation results and the drivers' subjective feedback, some refinements can be introduced to improve the realism and consistency of the scenario. In particular, the trigger timing of the first obstacle in Maneuver 1 should be revised so that the vehicle starts accelerating earlier, while in Maneuver 3 the simulation should avoid stopping immediately after the collision-avoidance action. These corrections would improve both the dynamic coherence of the scenario and the overall driver perception.